

8Byte.ai DevOps / SRE Assignment: Challenges & Resolutions

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Target Cloud: AWS (us-east-1)	Repository: github.com/sarthak24shirbhate/8byte-devops-assignment

1. Cyclic Dependency between ALB and ECS Security Groups

Problem:	Defining bidirectional least-privilege security group rules where ALB allows egress to ECS, and ECS allows ingress from ALB, caused cyclic dependency errors in Terraform's DAG.
Root Cause:	Inline ingress/egress blocks defined inside <code>aws_security_group</code> resources created mutual cross-dependencies preventing either group from determining its ID first.
Resolution:	Decoupled security groups from rules using standalone <code>aws_security_group_rule</code> resources so groups are created first, followed by cross-referencing rules.
Takeaway:	Always separate security group declarations from cross-referencing rules when implementing zero-trust perimeter boundaries in IaC.

2. S3 Remote State Backend Chicken-and-Egg Problem

Problem:	Terraform S3 backend configuration requires the remote S3 bucket and DynamoDB lock table to exist before terraform init can run, blocking automated single-command provisioning.
Root Cause:	Backend initialization occurs before any root module resources are evaluated or provisioned.
Resolution:	Created an isolated terraform/bootstrap/ module that provisions the encrypted, versioned S3 bucket and DynamoDB table independently prior to root state initialization.
Takeaway:	Decouple state storage lifecycles from application infrastructure to maintain clean GitOps workflows.

3. FastAPI TestClient Lifespan & SQLite Schema Initialization in Tests

Problem:	Running pytest on API integration endpoints failed with <code>OperationalError: no such table: items</code> .
Root Cause:	FastAPI lifespan handlers are not triggered by default when TestClient is instantiated outside of an ASGI context manager.
Resolution:	Implemented <code>tests/conftest.py</code> with an autouse session fixture (<code>setup_test_db</code>) and client context fixture to manage table creation and teardown independently.
Takeaway:	Test suites must manage test database schemas via standardized test fixtures rather than relying on application startup lifecycle hooks.

4. Trivy Package Name Issue in Python Requirements

Problem:	GitHub Actions failed during pip install with <code>'ERROR: No matching distribution found for trivy'</code> .
Root Cause:	Trivy is a standalone Go binary executed via aquasecurity/trivy-action, not a Python PyPI package.
Resolution:	Replaced trivy with pip-audit in requirements-dev.txt for dependency scanning and kept container image scanning handled by the Trivy GitHub Action in CI.
Takeaway:	Clearly delineate system-level binary tools (GitHub Actions) from language-specific dependencies (pip).

5. Securely Injecting RDS Passwords into ECS Fargate Tasks

Problem:	Containers need database credentials at launch, but embedding passwords in Git, env files, or .tfvars violates security policies.
Root Cause:	Passing database passwords via standard environment variables exposes them in task definitions and state outputs.
Resolution:	Generated credentials using random_password, stored payload in AWS Secrets Manager, and configured ECS task definition secrets array (valueFrom) with IAM Task Execution role permissions.
Takeaway:	Using AWS native IAM execution role resolution ensures credentials are retrieved in-memory at launch without persistent configuration exposure.

6. CloudWatch ELB Metric Dimension Mismatch

Problem:	CloudWatch ApplicationELB metrics displayed empty widgets because they require ARN Suffix strings rather than full ARNs.
Root Cause:	CloudWatch namespace dimensions for ELB strictly expect app// and targetgroup// strings.
Resolution:	Exported aws_lb.main.arn_suffix and aws_lb_target_group.app.arn_suffix from modules/alb/outputs.tf and wired them into modules/monitoring/main.tf widgets.
Takeaway:	Always inspect provider-specific metric dimension formats when constructing IaC monitoring dashboards.